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Inventor(s)	Neeb; Steffen et al.

DEVICE FOR LIFTING AND MOVING A STACK OF FLAT PRODUCTS LYING ON TOP OF ONE ANOTHER

Abstract

A gripping tool for raising and moving a stack of flat products lying on top of one another, such as folded products, is preferably a robot-guided gripping tool with a horizontally movable gripper for reaching under the stack. A first actuator positions the gripper in a first horizontal position or in a second horizontal position and a second actuator establishes a horizontal additional position, such as an intermediate position, for the gripper. The device can securely grip and handle stacks of different formats and to prevent collisions of the gripper, in particular gripper tines, with stacks, with other components, or with operating personnel. The device is suitable for use in the graphics industry and in particular in the environment of folding machines or their product delivery, where stacks of folded products lying on top of one another are stacked on pallets.

Inventors: **Neeb; Steffen (Bensheim, DE), Demir; Suat (Walldorf, DE), Östreicher; Michael (Eppelheim, DE), Leonhardt; Holger (Meckesheim, DE), Maier; Stefan (Dielheim, DE), März; Ralf (Angelbachtal, DE), Knöbl; Dominik (Angelbachtal, DE)**

Applicant: **Heidelberger Druckmaschinen AG** (Heidelberg, DE)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority, under 35 U.S.C. § 119, of German Patent Application DE 10 2024 105 246.5, filed Feb. 26, 2024; the prior application is herewith incorporated by reference in its entirety.

FIELD AND BACKGROUND OF THE INVENTION

[0002] The invention relates to a device for lifting and moving a stack of flat products lying on top of one another.

[0003] The invention lies in the technical field of the graphics industry and, there, in particular in the field of the handling (e.g. gripping, holding, moving, rotating, turning and/or depositing) of stacks of flexible, in particular printed and folded flat products, lying on top of one another, such as folded sheets, preferably made of paper, paperboard, cardboard, plastic or composite material, by way of a manipulator, in particular a jointed-arm robot/articulated-arm robot having a robot arm and having a gripper apparatus for the stacks.

[0004] Transporting stacks of planar products on a conveyor line, such as, for instance, a roller conveyor line, and using a gripper apparatus arranged on a robot arm to grasp the stacks, lift the stacks from the conveyor line, move the stacks to a pallet located near the conveyor line and deposit the stacks there is already known in the field of post press processing. See, for instance, German published patent application DE10 2020 103 398 A1 and its counterpart U.S. Pat. No. 11,584,605 B2 or German utility model DE 2020 19106975U1 and its counterpart U.S. Pat. No. 10,968,056 B2.

[0005] French Patent FR3110152B1 discloses a conveyor line and a robot-guided gripper head. The gripper head comprises a vertically movable upper gripping element, which is in the form of a clamping element, and a horizontally movable lower gripping element, which is in the form of a pair of tines.

[0006] German published patent application DE4124077A1 discloses an apparatus for receiving flexible planar parts. The apparatus comprises move-under elements which can be moved on a linear axis.

[0007] German published patent application DE4000262A1 discloses a gripping apparatus having tines which can be moved by means of a stroke rod and a pressure cylinder and which can be extended into a gripping position.

[0008] In practice, the problem that differently sized sheets, i.e. different formats, must be raised, moved and deposited as stacks often arises. An available gripper can be optimized for a specific format and can work less optimally in the case of other formats.

SUMMARY OF THE INVENTION

[0009] It is accordingly an object of the invention to provide a device which overcomes a variety of disadvantages associated with the heretofore-known devices and methods of this general type and which provides for an improvement over the prior art which, in particular, makes it possible to securely grip and handle stacks of different formats and to prevent collisions of the gripper, in particular gripper tines, with stacks, components or persons.

[0010] With the above and other objects in view there is provided, in accordance with the invention, a device for lifting, or raising, and moving a stack of flat products lying on top of one another. The device comprises: [0011] a horizontally movable gripper configured for reaching under the stack; [0012] a first actuator configured for selectively positioning said gripper in a first horizontal position or in a second horizontal position; and [0013] a second actuator configured for establishing an additional horizontal position for said gripper.

[0014] In other words, the above and other objects are achieved, according to the invention, by a device, in particular a gripping tool and preferably a robot-guided gripping tool, for lifting and moving a stack of flat products lying on top of one another, e.g. folded products. The novel device comprises a horizontally movable gripper for reaching under the stack and a first actuator which positions the gripper in a first horizontal position or in a second horizontal position. The device further comprises a second actuator which is different from the first actuator and which establishes a horizontal additional position, preferably an intermediate position, for the gripper.

[0015] The invention advantageously makes it possible to securely grip and handle stacks of different formats and to prevent collisions of the gripper, in particular gripper tines, with stacks, components or persons.

[0016] The invention is used preferably in the graphics industry and in particular in the environment of folding machines or their product delivery, where e.g. stacks of folded products lying on top of one another are stacked on pallets.

[0017] A primary advantage of the invention is that the gripper, and thus also its (projecting) tines can—from its position largely retracted into a housing, or parking position or non-working position—be moved into and positioned in not only a single (gripping or working) position but at least two mutually different (gripping or working) positions, one of which is referred to as the additional position and in particular as intermediate position. The intermediate position, such as, for example, a half-extended position, can be used in particular for gripping and holding smaller formats, while in the completely extended position preferably large formats are handled. This results in the advantage that, in the case of smaller formats, no tines protrude (far) beyond the gripped stack and collide, during depositing, with stacks already previously deposited on pallets and shift or knock over said stacks; risks of injury to the operating personnel which result from the protruding can also be prevented in this way.

[0018] The robot arm for guiding the gripping tool is preferably part of a robot system, which usually also comprises a robot base and which can be in the form of a typical industrial robot together with fencing for operator protection, in particular in the form of a jointed-arm robot having three to seven axes of rotation, or in the form of a so-called collaborative robot system, in particular in the form of a so-called cobot (which is short for collaborative robot). The latter does not require any fencing, because the system has its own sensor equipment and uses the sensor equipment to sense contact with operating personnel and to prevent possible injuries, e.g. by means of an automatic stop of the robot arm movement. A collaborative robot system can also be realized by means of an industrial robot with additional region scanner (sensing of operating personnel in the hazard region) and with an automatic stop system, as an alternative to the cobot.

[0019] Below, preferred developments or variations of the invention (“developments” for short) are described. Where technically possible or feasible, the variants may be selectively combined with one another.

[0020] In accordance with one development of the invention, the second actuator establishes the additional position by performing a shifting movement.

[0021] A further development can be characterized in that the second actuator shifts a movable stop for the gripper or that the shifting movement moves the stop. One development can be characterized in that the stop is mounted for rotation. One development can be characterized in that the second actuator positions the stop in a passive position or in at least one active position. In the active position, the stop can act as a stop for the gripper and thereby position, in particular stop, the movable gripper in the additional position during movement of the gripper. One development can be characterized in that the stop is coupled to the second actuator by a linkage.

[0022] In accordance with another development, during retraction, the first actuator and/or the

gripper stops at the stop in its active position during extension. One development can be characterized in that, during extension, the first actuator and/or the gripper moves the stop out of its active position, e.g. carries the stop along.
[0023] According to a further feature of the invention, the second actuator is preferably a pneumatic cylinder or alternatively a hydraulic cylinder or an electric cylinder.
[0024] In accordance with yet a further development the device comprises at least one additional actuator, which establishes a respective other horizontal additional position for the gripper.
[0025] In accordance with yet another development can the gripper is arranged on the device for movement relative to a frame and/or housing of the device, in particular for retraction and extension.
[0026] It should be understood that the features and combinations of features disclosed in the above descriptions of developments and in the following description of exemplary embodiments may be combined with one another to achieve additional and advantageous developments of the invention.
[0027] As an alternative, the objects of the invention may be achieved in that the second actuator shifts the gripper (into at least two positions; relative to the first actuator) or the first actuator (into at least two positions; together with the gripper) and thereby establishes the horizontal additional position for the gripper.
[0028] As another alternative, the objects can be achieved in that the first actuator establishes the horizontal additional position for the gripper, e.g. in that it moves the gripper into the additional position. In that case, the second actuator can be omitted. The first actuator can be in the form of a —preferably pneumatic—multi-position cylinder.
[0029] As yet another alternative, the objects of the invention can be achieved in that the gripper comprises movable elements, e.g. tines, which can be moved into the additional position by means of an actuator.
[0030] Other features which are considered as characteristic for the invention are set forth in the appended claims.
[0031] Although the invention is illustrated and described herein as embodied in a device for raising and moving a stack of flat products lying on top of one another, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims.
[0032] The construction and method of operation of the invention, however, together with additional objects and advantages thereof will be best understood from the following description of specific embodiments when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0033] FIGS. **1** to **4** show preferred exemplary embodiments of the invention and of the developments. Corresponding features are identified with the same reference signs throughout the figures. Certain reference signs for elements that are repeatedly shown in the figures have been omitted in some cases for the sake of clarity.
[0034] FIG. **1** shows a schematic plan-view illustration of a conveyor line of a machine with a preferred embodiment of the device according to the invention.
[0035] FIG. **2** shows a sectional side view of a preferred embodiment of the device according to the invention with a stop in a passive position.
[0036] FIG. **3** shows a sectional side view of the same preferred embodiment of the device according to the invention with the stop in an active position.
[0037] FIG. **4** shows a sectional side view of another preferred embodiment of the device according

to the invention with two stops.

DETAILED DESCRIPTION OF THE INVENTION

[0038] Referring now to the figures of the drawing in detail and first, in particular, to FIG. **1** thereof, there is shown a portion of a machine **1** of the graphics industry, preferably a post press processing machine **1** and in particular a folding machine **1**, and a preferably linear conveyor line **2** downstream of the machine **1**, for example a roller table **2** having transport rollers, with a conveying direction **3** and with at least one motor **4** for preferably horizontally conveying successive stacks **5** of products lying on top of one another, for example printed and folded sheets of paper. The machine **1** and the conveyor line **2** are arranged on a floor **80** of a production facility, for example a print shop, and can be operated by an operating person **81**.

[0039] FIG. **1** also shows an apparatus for transferring the stacks **5**, in particular for depositing the stacks **5** on pallets **8**. The apparatus comprises a robot **10** having a movable robot arm **11**, for example a jointed arm having a plurality of joints **12**, on which robot arm a gripping tool **20** for gripping individual stacks **5** is arranged. A hazard region **13** is defined by the maximum movements of the robot arm **11**. The gripping tool **20** grips one stack **5** at a time, lifts the stack from the conveyor line **2** and holds the stack **5** during the robot movement **14** to a pallet **8**. The stacks **5** conveyed away from the machine **1** are in this way deposited one after another on the pallet **8** and form there (arranged next to one another and on top of one another according to a specified depositing scheme) a pallet stack. By way of example, in FIG. **1** two pallets **8** are placed on a longitudinal side of the conveyor line. The robot **10** is also placed on the longitudinal side of the conveyor line, by way of example; the robot could alternatively also be arranged at the end of the conveyor line **2**. A controller **15** is provided which controls the movements **14** of the robot arm **11** and the movements of the grippers of the gripping tool **20** during the transfer of the stacks **5**. An operator **81** can remove stacks **5** at the end of the conveyor line **2** for inspection purposes. Details of the gripping tool **20** are presented in the following FIGS. **2** to **4**.

[0040] FIGS. **2** and **3** show a device **20**, in particular a robot-guided gripping tool, comprising a housing **21**, a horizontally (see orientation **90**) movable gripper **22** and a vertically (see orientation **91**) movable gripper **23** (with actuator). For the horizontal moving of the gripper **22** (relative to the housing **21**), a first actuator **30** is connected to the gripper **22**. The first actuator **30** makes it possible to move the gripper **22** back and forth from a first position **40** to a second position **41** or to position the gripper **22** in the two positions **40** and **41**. The first position **40** is a position in which the gripper **22** grips and handles large stacks **5***a* (first actuator **30** retracted), and the second position **41** is a position in which the gripper **22** is completely or nearly completely retracted into the housing **21** (first actuator **30** fully extended).

[0041] According to the invention, the device **20** comprises a second actuator **31**, preferably a pneumatic cylinder, which establishes or defines an additional position **42**, in particular an intermediate position **42**. In the additional position **42**, smaller stacks **5***b* can be gripped and handled (the terms "large" and "small" are to be understood in relation to one another). For this purpose a stop **33**, which is connected to the second actuator **31** by a linkage **35**, is moved back and forth (see shifting movement **44** of the second actuator **31**) between its passive position **45** (see FIG. **2**) and its active position **46** (see FIG. **3**) or positioned in one of the two positions **45** or **46**. The linkage **35** in turn can be mounted for rotation about a pivot point **37**. If the stop **33** is in the passive position **45**, the gripper **22** can—as described above—be moved back and forth. If, in contrast, the stop is in the active position **46**, it forms a stop for the gripper **22** in the intermediate position **42**, i.e. the gripper **22** moved by the first actuator **30** can be moved back and forth only between the second position **41** and the additional position **42**. If the stop **33** is moved back into or positioned in its passive position, the gripper **22** can again be moved back and forth between the first position **40** and the second position **41**.

[0042] FIG. **4** shows that, besides the second actuator **31**, at least one additional actuator **32** can be provided. The additional actuator **32** can move an additional stop **34** back and forth between its

passive position and its active position or position said additional stop. The additional stop **34** can be connected to the additional actuator **32** by an additional linkage **36** and, for example, can be mounted for rotation about the same pivot point **37**. This makes it possible to establish or define two different additional positions **42**, for example two different intermediate positions, by means of the stop **33** and the additional stop **34**. This in turn means that the gripper **22** can be moved back and forth between the second position **41** and one of the two additional positions **42**. While the active positions of the two stops **33** and **34** differ from one another, the passive positions can be the same. The two active positions differ at least with respect to their position in the horizontal direction **90**.
[0043] The following is a summary list of reference numerals and signs and the corresponding structure used in the above description of the invention: [0044] **1** Machine, in particular folding machine [0045] **2** Conveyor line, in particular roller table [0046] **3** Conveying direction [0047] **4** Motor/motors [0048] **5** Stack of flat products lying on top of one another [0049] **5***a* Large stack (large format) [0050] **5***b* Small stack (small format) [0051] **6** Receiving position, at the conveyor line [0052] **7** Depositing position, at the pallet [0053] **8** Pallet/pallets [0054] **9** Pallet stack [0055] **10** Robot/robot system [0056] **11** Robot arm [0057] **12** Joints [0058] **13** Hazard region [0059] **14** (Depositing) movement [0060] **15** Controller [0061] **20** Device, in particular gripping tool [0062] **21** Frame or housing [0063] **22** Gripper (horizontally movable) [0064] **23** Gripper (vertically movable) with actuator [0065] **30** First actuator [0066] **31** Second actuator [0067] **32** Additional actuator [0068] **33** Stop [0069] **34** Additional stop [0070] **35** Linkage [0071] **36** Additional linkage [0072] **37** Pivot point [0073] **40** First position (gripping position for large stack) [0074] **41** Second position (gripper completely retracted) [0075] **42** Additional position, in particular intermediate position (gripping position for small stack) [0076] **43** Other additional position, in particular intermediate position (gripping position for medium stack) [0077] **44** Shifting movement of actuator [0078] **45** Passive position of the stop [0079] **46** Active position of the stop [0080] **47** Shifting movement of gripper [0081] **80** Floor of a production facility [0082] **81** Operating person [0083] **90** Horizontal [0084] **91** Vertical

Claims

1. A device for lifting and moving a stack of flat products lying on top of one another, the device comprising: a horizontally movable gripper configured for reaching under the stack; a first actuator configured for selectively positioning said gripper in a first horizontal position or in a second horizontal position; and a second actuator configured for establishing an additional horizontal position for said gripper.
2. The device according to claim 1, wherein said second actuator is configured to establish the additional horizontal position by performing a shifting movement.
3. The device according to claim 1, which comprises a movable stop for said gripper, and wherein said second actuator is configured to shift said movable stop for said gripper.
4. The device according to claim 3, wherein said movable stop is mounted for rotation.
5. The device according to claim 3, wherein said second actuator is configured to selectively position said movable stop in a passive position or in at least one active position.
6. The device according to claim 3, which comprises a linkage coupling said movable stop to said second actuator.
7. The device according to claim 3, wherein, during a retraction, said first actuator stops at said movable stop in an active position thereof.
8. The device according to claim 7, wherein, during an extension, said first actuator moves said movable stop out of the active position.
9. The device according to claim 1, wherein said second actuator is a cylinder device selected from the group consisting of a pneumatic cylinder, a hydraulic cylinder, and an electric cylinder.

10. The device according to claim 1, further comprising at least one additional actuator configured for establishing another additional horizontal position for said gripper.

11. The device according to claim 1, further comprising at least one of a frame or a housing, and wherein said gripper is arranged for movement relative to said frame and/or said housing of the device.